TOPIC 17/60

What is Top-K in LLMs?

> Cleaning up the AI's output.
>
> How we filter the "noise" of bad ideas.

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THE PROBLEM

When an LLM predicts the next word, it assigns a probability to **every word** in its entire dictionary.

Prompt: "The sky is..."

Blue (90%)

Clear (7%)

Dark (2%)

...Spaghetti (0.0001%)

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TOP-K DEFINED

Truncating the tail

Top-K sampling simply **chops off** the bottom of the list. We only keep the top "K" most probable words.

If K=50: The AI ignores everything outside the top 50 choices. It effectively kills the "weird" words like "Spaghetti".

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THE BENEFIT

Quality Control

Models often have a "long tail" of low-probability words that are technically possible but nonsensical in context.

Without Top-K:

The AI might pick a word with 0.001% probability, leading to sudden, nonsensical gibberish mid-sentence.

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CONFIGURING THE MODEL

The K-Value

The Size of the Pool

The value of 'K' determines how "adventurous" the AI is allowed to be.

> **K = 1:** Deterministic (The AI always picks the #1 word). Boring, but safe.
>
> **K = 50-100:** Creative/Natural. Good balance of logic and variety.

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THE PAIR

Temp + Top-K

These settings work together as a 1-2 punch to control LLM output.

The Workflow:

1. Temperature scales the probabilities.
2. Top-K chops off the tail.

> Result: Only the "good" candidates remain.

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VISUALIZING

NO TOP-K (TAIL RISK)

The AI samples from the entire dictionary.

It might accidentally pick from the "junk" words at the very end of the list.

WITH TOP-K

The AI samples from a refined pool.

Every option in the pool is high-quality, so no matter what it picks, the output remains coherent.

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THE LIMITATION

Static K

A static "K" is rigid. Sometimes there are 100 great words, sometimes there are only 2.

If K is too small, you limit the model's creativity.

This is why modern models often use **Top-P (Nucleus) Sampling** instead, which dynamically adjusts the K-value based on confidence.

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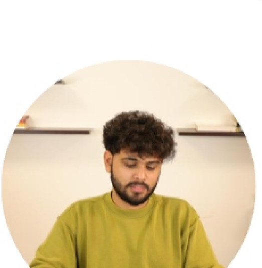

WHY IT MATTERS

Consistency

Without Top-K, AI output can be wildly unpredictable and erratic.

Reliability

It forces the model to stay "on the rails," resulting in much higher quality writing.

User Experience

Crucial for creative writing and code, where one word choice can change the entire result.

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Top-K Sampling

CONCEPT UNLOCKED

- 🎯 Filters by rank (Top K)
- ✂️ Chops off the "junk" tail
- 🛡️ Prevents hallucinations
- 📈 Ensures high-quality output

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